

b_i = vector of risk exposures
 b_{ij} = risk sensitivity of stock i to factor j
 e_i = vector of errors (unexplained return)
 e_{it} = error term of stock i in period t
 n = total number of time periods
 m = total number of stocks
 k = total number of factors

Constructing Factor Independence

Real-world data often results in factors that are not independent which violates regression property (5). This makes it extremely difficult to determine accurate risk exposures to these factors. In these situations, analysts can transform the set of dependent original factors into a new set of factors that are linearly independent (Kennedy, 1988).

This process is described as follows:

Let F, G, and H represent three explanatory factors that are correlated.

First, sort the factors by explanatory power. Let F be the primary driver of risk and return, let G be the secondary driver, and let H be the tertiary driver.

Second, remove the correlation between F and G. This is accomplished by regressing the secondary factor G on the primary factor F as follows:

$$G = \tilde{\nu}_0 + \tilde{\nu}_1 F + e_G$$

The error term in this regression e_G is the residual factor G that is not explained by the regression model and by definition (property (4)) is independent of F. Then let $\tilde{G}$ be simply e_G from the regression. That is:

$$\tilde{G} = G - \tilde{\nu}_0 - \tilde{\nu}_1 F$$

Third, remove the correlation between factor H and factor F and the new secondary factor $\tilde{G}$. This is accomplished by regressing H on F and $\tilde{G}$ as follows:

$$H = \hat{\gamma}_0 + \hat{\gamma}_1 F + \hat{\gamma}_2 \tilde{G} + e_H$$

The error term in this regression e_H is the residual factor H that is not explained by the regression model and by definition (property (4)) is independent of F and G. This process can be repeated for as many factors as are present.